Additional Exercises 4

Taylor Approximations & Uniform Convergence

Mirrors: SIT4 | ★ Easy ★★ Medium ★★★ Hard

Exercise 1: Numerical approximation via Taylor polynomials

Use the Taylor polynomial $P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$ *and the Lagrange remainder* $|R_n| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$ *to achieve the stated accuracy.*

(a) ★ Approximate $e^{0.1}$ with error $< 10^{-4}$. How many terms of the Maclaurin series are needed?

(b) ★ Approximate $\sin(0.2)$ with error $< 10^{-5}$.

(c) ★ Approximate $\ln(1.1)$ with error $< 10^{-3}$.

(d) ★★ Approximate $\sqrt[3]{28}$ using $f(x) = (27+x)^{1/3}$ expanded around $a = 0$. Achieve accuracy < 0.001.

(e) ★★ Approximate $\cos(5^\circ)$ (in radians: $x = \pi/36$) with error $< 10^{-6}$.

(f) ★★ Using the series for $\arctan x$, approximate $\pi/4$ with error $< 10^{-3}$. How many terms are needed?

(g) ★★★ Using $\ln 2 = \ln \frac{3}{2} - \ln \frac{3}{4}$ and known series, approximate $\ln 2$ with error < 0.001. Compare efficiency with the direct $\ln(1+x)$ series.

(h) ★★★ Approximate $\displaystyle\int_0^{0.5} e^{-x^2}\,dx$ with error $< 10^{-4}$ by integrating the Maclaurin series term by term.

Exercise 2: Uniform convergence of sequences of functions

A sequence $f_n \to f$ *uniformly on* E *iff* $\sup_{x\in E}|f_n(x) - f(x)| \to 0$. *Pointwise convergence alone does NOT imply continuity, integrability, or differentiability of the limit.*

(a) ★ $f_n(x) = x^n$ on $[0,1)$. Find the pointwise limit. Is convergence uniform?

(b) ★ $f_n(x) = \dfrac{x}{n}$ on $\mathbb{R}$. Determine pointwise and uniform convergence.

(c) ★ $f_n(x) = \dfrac{1}{nx+1}$ on $[1, \infty)$. Show convergence is uniform.

(d) ★★ $f_n(x) = \dfrac{nx}{1+n^2x^2}$ on $[0, 1]$. Find pointwise limit and determine whether convergence is uniform.

(e) ★★ $f_n(x) = n^2 x\, e^{-nx}$ on $[0, 1]$. Show $f_n \to 0$ pointwise but $\int_0^1 f_n\, dx \not\to 0$. What does this tell you about the convergence?

(f) ★★ $f_n(x) = \dfrac{\sin(nx)}{\sqrt{n}}$ on $[0, 2\pi]$. Show the sequence converges uniformly. Does it converge in C^1?

(g) ★★★ $f_n(x) = x^n(1-x)$ on $[0, 1]$. Find the pointwise limit and show convergence is uniform on $[0, 1]$. *(Hint: find $\sup_{x\in[0,1]} |f_n(x)|$ analytically.)*

(h) ★★★ Show that if $f_n \to f$ uniformly on $[a, b]$ and each f_n is continuous, then $\lim_{n\to\infty} \int_a^b f_n(x)\, dx = \int_a^b f(x)\, dx$.

Exercise 3: Uniform convergence of series of functions

Weierstrass M-test: if $|f_n(x)| \le M_n$ on E and $\sum M_n < \infty$, then $\sum f_n(x)$ converges uniformly (and absolutely) on E.

(a) ★ Show $\displaystyle\sum_{n=1}^{\infty} \frac{\cos(nx)}{n^2}$ converges uniformly on $\mathbb{R}$.

(b) ★ Show $\displaystyle\sum_{n=0}^{\infty} x^n$ converges uniformly on $[-r, r]$ for any $0 < r < 1$, but not on $(-1, 1)$.

(c) ★★ Show $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{n}$ converges uniformly on $[-1, 1]$. *(Use Dirichlet/Abel test for uniform convergence.)*

(d) ★★ Determine where $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n^2+x^2}$ converges uniformly.

(e) ★★ Let $S(x) = \displaystyle\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^3}$. Show S is continuous and differentiable, and find $S'(x)$.

(f) ★★★ Let $f(x) = \displaystyle\sum_{n=0}^{\infty} (-1)^n x^{2n}$ for $|x| < 1$. Show the series converges uniformly on $[-r, r]$ for $r < 1$, identify the sum, and verify by differentiating term by term.

(g) ★★★ Show that a power series $\displaystyle\sum_{n=0}^{\infty} a_n x^n$ with radius of convergence R converges uniformly on $[-r, r]$ for any $r < R$. Why does it generally fail to converge uniformly on $(-R, R)$?

★ *Taylor remainder formula, sup-norm definition*

★★ *Weierstrass M-test, uniform convergence criteria, term-by-term integration/differentiation*

★★★ *Abstract convergence theorems, subtle counterexamples.*